

Adaptive Confidence-weighted Expansion for Trustworthy Multi-Omics Multimodal Fusion

Mohammad Raahemi¹ · Ali Sekhavati¹ · Alireza Maleki² · Hamid Nasiri^{3,*}

¹ Faculty of Engineering, University of Ottawa, Ottawa, Canada

² RIV Lab, Department of Computer Engineering, Bu-Ali Sina University, Hamedan, Iran

³ School of Computing and Communications, Lancaster University, Lancaster, UK

Trustworthy multimodal learning · Multi-omics classification · Confidence calibration



KEY TAKEAWAY

ACE expands each omics modality using intra-modality correlations, learns confidence at both modality and fused-decision levels, and produces more accurate, better calibrated predictions under noisy or partially uninformative inputs.

4
datasets

2n
expanded views

Clinical motivation

Multimodal learning can improve medical prognosis, but noisy, missing, or uninformative modalities can make a model confidently wrong. This risk is amplified in multi-omics cohorts, where the number of molecular features is large while the number of patients is comparatively small.

The gap

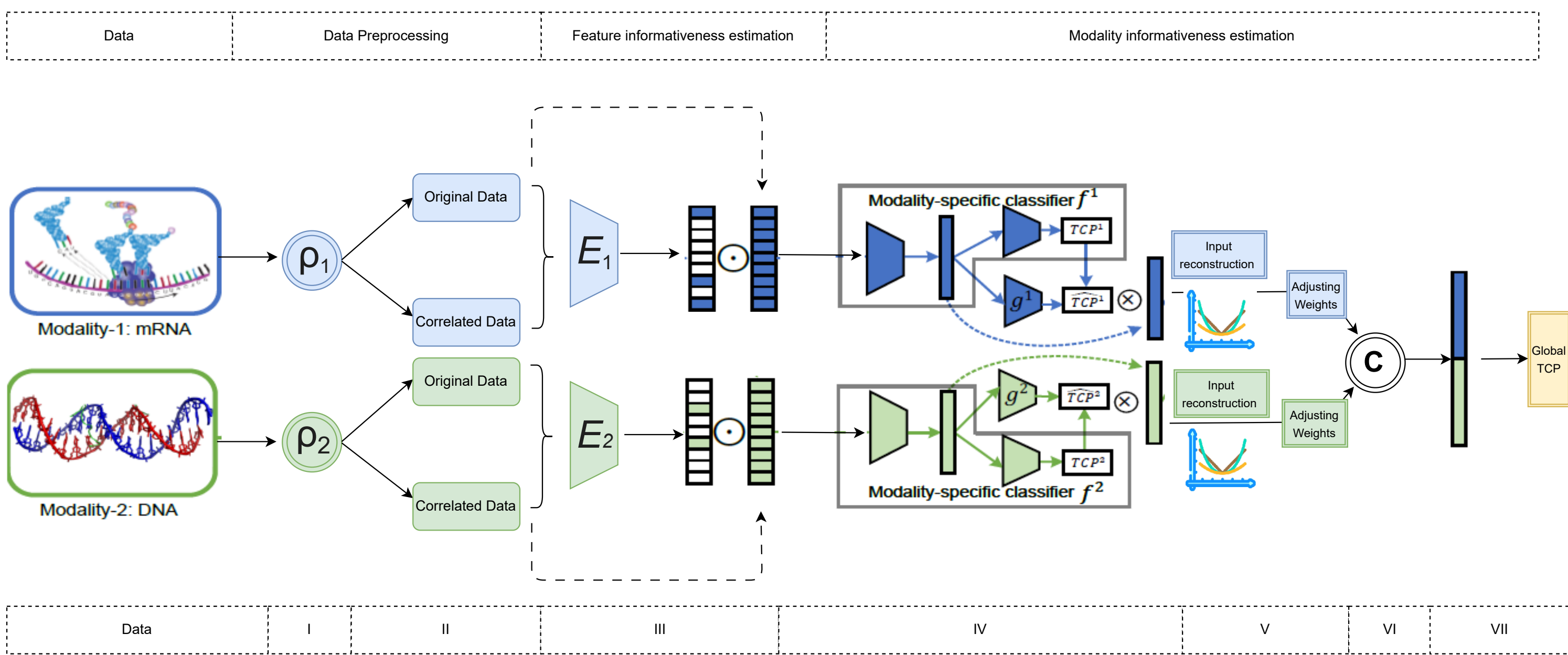
Conventional early, intermediate, and decision-level fusion usually treat modality quality as fixed. They rarely identify which view is unreliable for a specific patient while also reporting whether the final fused decision itself should be trusted.

ACE contributes

- ▶ Correlation-based expansion that exposes complementary intra-omics structure.
- ▶ Adaptive confidence weighting that suppresses unreliable views before fusion.
- ▶ Modality-specific and global TCP heads for local and end-to-end trust estimation.
- ▶ Input reconstruction and noise injection for stable learning from limited data.

ACE architecture

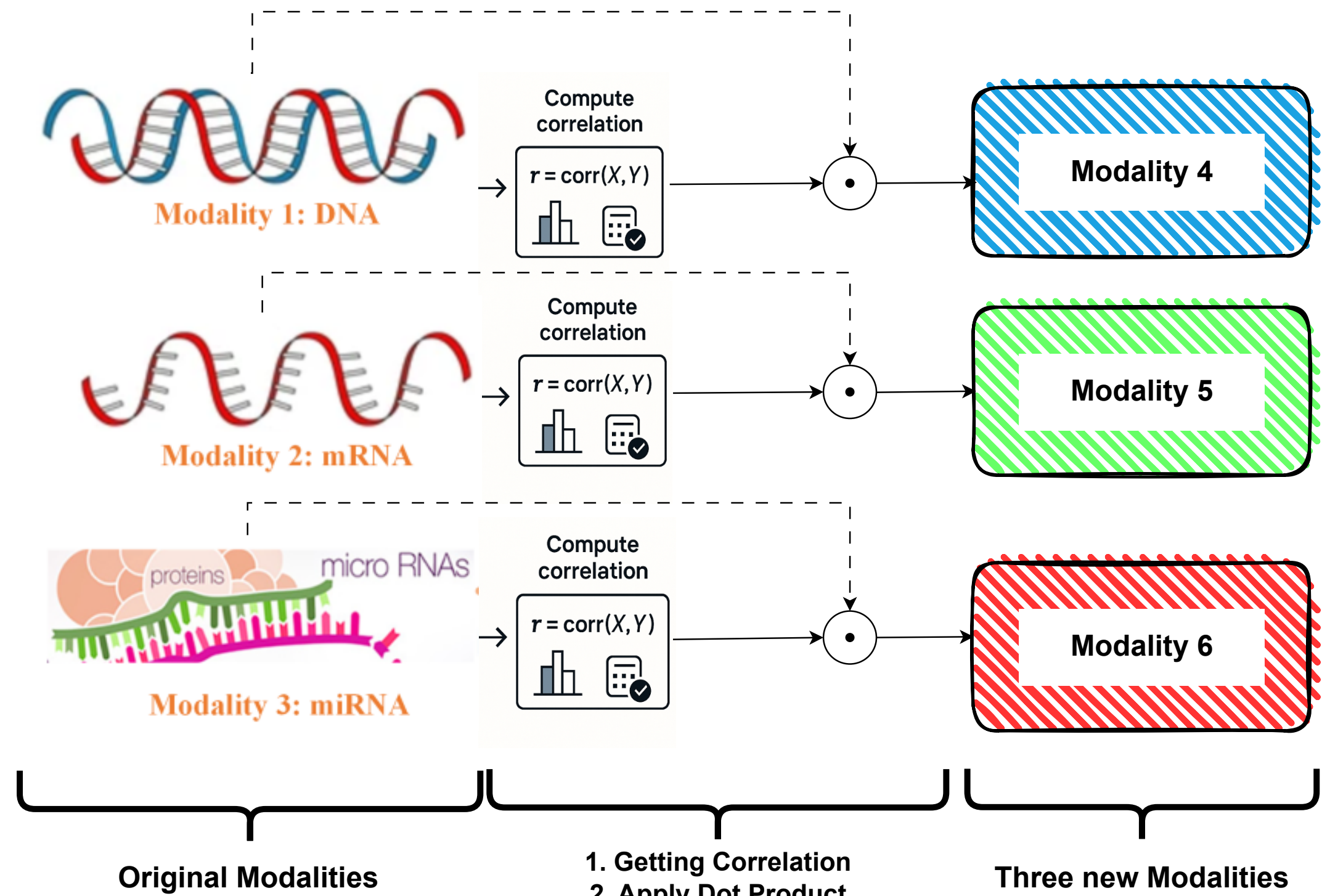
Original + correlated views, gated encoders, calibrated modality weighting, global trust



Pipeline logic

Expand: generate a correlated companion view for each modality. **Calibrate:** gate features, reconstruct inputs, and estimate per-view confidence. **Fuse & trust:** scale latent features by confidence, concatenate them, and predict a global TCP score.

Correlation-driven modality expansion



Core transformation

For modality i :

$$\mathbf{X}'_i = \mathbf{X}_i \text{corr}(\mathbf{X}_i)$$

Correlation is estimated only from training data to prevent information leakage.

Why it helps

- ▶ Re-expresses each feature as a correlation-weighted combination of related features.
- ▶ Emphasizes stable co-expression patterns while reducing weak or noisy dimensions.
- ▶ Doubles n original modalities into $2n$ complementary views.

Important

The transformed view complements rather than replaces the original signal. The same training-derived correlation matrix is applied to test samples, preventing information leakage.

Learning objective

Confidence-weighted fusion and loss

$$\hat{\mathbf{h}}_i = \alpha_i \mathbf{h}_i, \quad \mathcal{L}_{aw,i} = \lambda_{aw} \sigma(\alpha_i) \text{CE}(\mathbf{z}_i, \mathbf{y})$$

The same learned confidence both attenuates unreliable latent features before fusion and controls their contribution during optimization.

Self-supervised reconstruction

$$\mathcal{L}_{rec,i} = \lambda_{rec} \|\mathbf{X}_i - \mathbf{g}_i(\mathbf{h}_i)\|_2^2$$

Recovering clean inputs from corrupted views forces each encoder to retain modality structure and improves robustness to noise and missing values.

Global TCP

A dedicated confidence head operates on the fused embedding. It captures cross-modal agreement and conflict that cannot be observed from modality-specific confidence estimates alone.

Joint training objective

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{global}} + \sum_i (\mathcal{L}_{\text{TCP},i} + \mathcal{L}_{\text{rec},i} + \mathcal{L}_{\text{aw},i} + \mathcal{L}_{\text{feat},i})$$

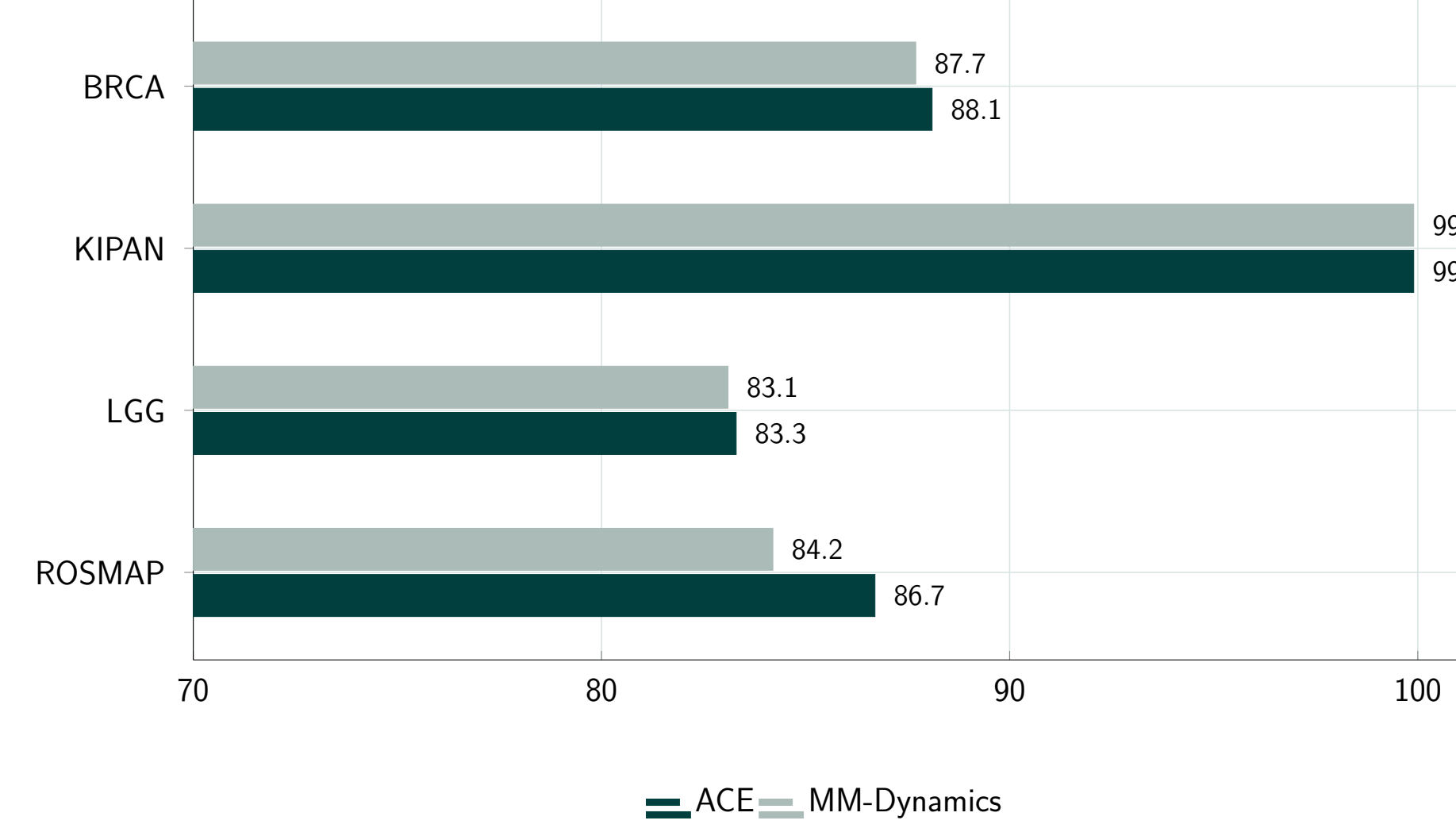
Classification, calibration, reconstruction, adaptive weighting, and sparse feature selection are optimized end to end.

Experiments & Results

Note: Results are averaged over 20 independent runs on BRCA · KIPAN · LGG · ROSMAP

Predictive performance

Accuracy comparison: ACE vs. MM-Dynamics



ROSMAP accuracy

+2.5 pp

86.7% vs 84.2%

ROSMAP F1

+2.2 pp

86.8% vs 84.6%

ACE also reaches 88.1% BRCA accuracy and 99.9% KIPAN accuracy, while preserving strong LGG performance. The largest practical gain appears on ROSMAP, a noisy clinical cohort where reliability-aware fusion is especially valuable.

Evaluation protocol

BRCA and KIPAN are evaluated with accuracy, weighted F1, and macro F1; LGG and ROSMAP use accuracy, F1, and AUC. Every experiment is repeated 20 times with different random seeds and mean performance is reported.

Confidence calibration

Direct per-view comparison with MM-Dynamics

BRCA View 0 MSE

72%

0.0472 vs 0.1669

BRCA AUROC-Error

0.843

vs 0.506 baseline

ROSMAP View 2 ECE

30%

0.1910 vs 0.2740

ROSMAP View 0 NLL

42%

0.5257 vs 0.9002

Statistical significance

- ▶ BRCA: all three metrics significant ($p = 0.0017$ – 0.0034).
- ▶ ROSMAP: all metrics significant ($p < 0.0001$).
- ▶ KIPAN: accuracy and weighted F1 significant ($p = 0.0375$).
- ▶ LGG: AUC significant ($p < 0.0001$).

How to read the calibration results

Lower ECE, MCE, NLL, and MSE mean predicted confidence is more aligned with correctness. Higher AUROC-Error and AUPRC-Error mean the confidence signal is more useful for identifying likely mistakes.

Across both original and correlation-derived views, ACE reduces overconfidence and provides a stronger basis for abstention, human review, or additional testing in high-stakes clinical workflows.

Ablation & conclusion

Largest performance drop

BRCA full ACE

88.12%

No reconstruction + noise

85.53%

ROSMAP full ACE

86.65%

No reconstruction + noise

79.86%

What the ablation shows

- ▶ Input reconstruction + noise injection is the strongest generalization component.
- ▶ Correlation expansion and adaptive weighting improve robustness and fusion stability.
- ▶ Global TCP mainly refines trust calibration, with limited impact on raw accuracy.

Interpretation and next steps

Reconstruction protects against data scarcity, confidence weighting limits propagation of weak modalities, and global TCP supplies an interpretable sample-level trust signal. Future work will test hyperparameter sensitivity and extend ACE to medical imaging and other data types.

Bottom line

ACE offers a practical route to multimodal predictions that are both more accurate and more trustworthy for safety-critical clinical decision support.

Selected references

- [1] Z. Han et al., "Multimodal Dynamics: Dynamical Fusion for Trustworthy Multimodal Classification," CVPR, 2022.
- [2] T. Wang et al., "MOCONET: Multi-omics Integration via Graph Convolutional Networks," Nature Communications, 2021.
- [3] Z. Han et al., "Trusted Multi-view Classification with Dynamic Evidential Fusion," IEEE TPAMI, 2022.